

SIGNOVA — Project Plan

Emerging Trends (Emergex), SAIT · Sep 27, 2026 · @Vansh Tuteja

SIGNOVA is a working prototype that turns spoken English into hand-shape signs on a 3D-printed robotic hand: Speech → AI → Sign Representation → Robotic Motion. It demonstrates the pipeline; it does not claim to be a sign-language translator.

Overview

Problem. People who speak and people who sign often lack a shared channel in everyday moments. SIGNOVA explores whether AI can leave the screen and communicate through physical motion.

Goals for prototype v1

- 1. Hear a spoken sentence and produce an ASL-style gloss (sign order) from a fixed, validated vocabulary.
- 2. Perform each sign on the robotic hand, fingerspelling anything outside the vocabulary.
- 3. Show every stage live on a dashboard: transcript, gloss, current sign, hand simulation.
- 4. Have the signs checked by a fluent ASL user and report the results honestly.

Success looks like: a fluent signer recognises at least 80% of Tier 1 signs, the hand starts moving within 3 seconds of the speaker stopping, and the demo runs 20 times in a row without a hardware fault.

Course themes covered: Physical AI and robotics, Generative AI/NLP, AI-assisted development, accessibility and inclusive design, responsible AI, interdisciplinary skills.

Scope

v1 covers one right hand doing static handshapes and a few simple motions. Everything else is documented as future work.

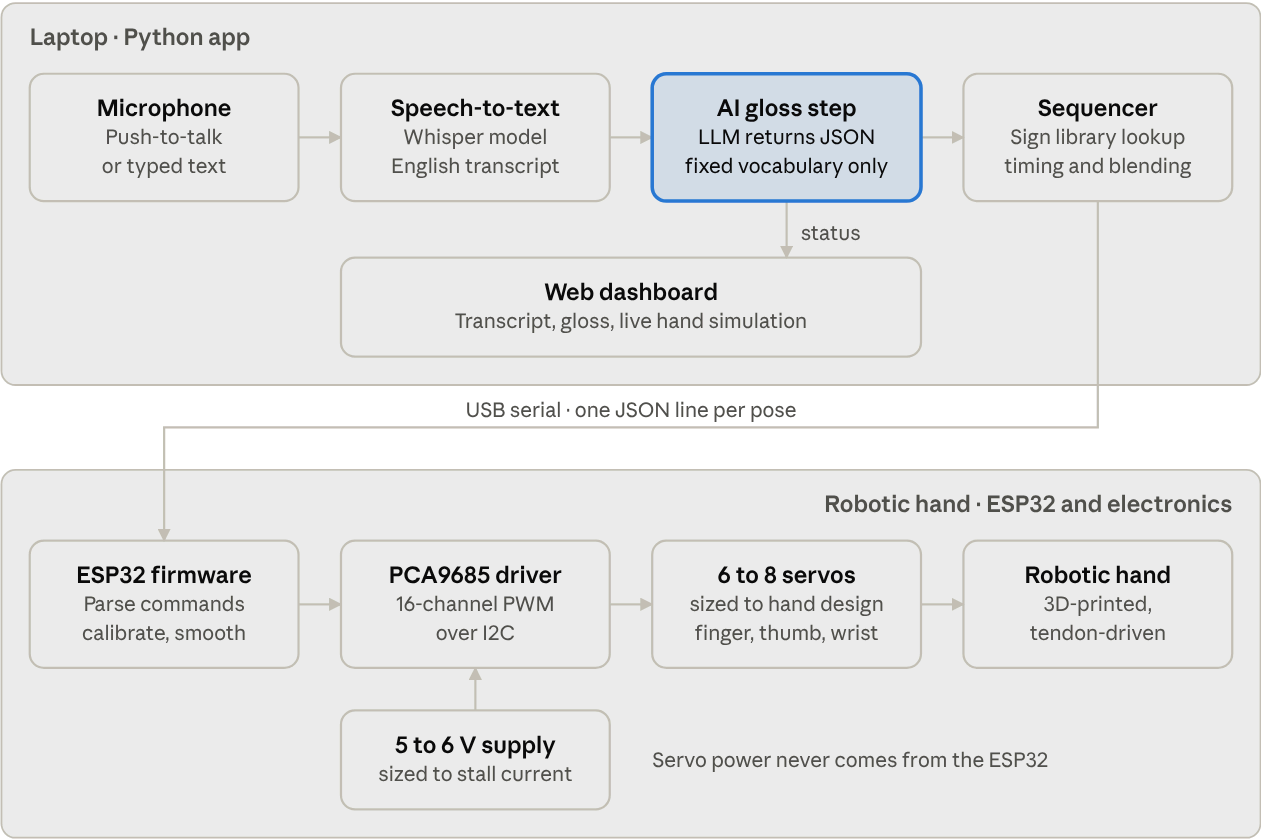
In scope (v1)	Out of scope (future work)
One right hand, 5 fingers + thumb rotation (+ wrist if hardware allows)	Two hands, full arm, elbow and shoulder movement

In scope (v1)	Out of scope (future work)
Static handshapes: ASL numbers, selected letters, "I love you"	Facial expressions and other non-manual markers, which carry ASL grammar
Fingerspelling of short words using supported letters	Full ASL grammar, classifiers, spatial referencing
English speech input in a quiet room + typed fallback	Other spoken languages, noisy environments
ASL only	LSQ, BSL and regional variations
Laptop + ESP32 over USB	Standalone device, mobile app

System architecture

All speech, AI and sequencing run on a laptop; the ESP32 only receives finger positions and drives servos. This keeps the firmware small and lets the software team work without the hand.

The laptop does the thinking; the ESP32 only moves servos



SIGNOVA architecture · laptop pipeline to ESP32 and hand

Component	Responsibility	Owner
Speech-to-text	Turn audio into an English transcript	Software team
AI gloss step	Map the transcript to supported signs + fingerspelled words, in ASL-style order	Software team
Sequencer	Look up each sign's keyframes, add transitions and hold times, send poses	Software team
Dashboard + simulator	Show every stage live; render a virtual hand from the same pose data	Software team
ESP32 firmware	Parse pose commands, apply calibration, move servos smoothly	Software team writes, hardware team tunes

Component	Responsibility	Owner
Hand, servos, power	Mechanical build, wiring, calibration values	Hardware (arm) lead

Hardware plan

Recommendation: start from the open-source InMoov hand, add a second thumb servo and wrist rotation, and drive everything from an ESP32 through a PCA9685 board on its own power supply. The number of joints decides which signs are possible, so this is the first decision to lock.

Candidate hand designs

Design	Joints	Actuation	Fit for SIGNOVA
InMoov hand + forearm	1 bend per finger, wrist rotation, optional 2nd thumb servo	Braided-line tendons (10 × 75 cm), standard-size servos (HK15298 or MG946R), MG996 on the wrist	Best documented and easiest to print. No finger spread, so U and V look the same.
PARLOMA hand (Bulgarelli et al., 2016)	8 DOF + 3-DOF spherical wrist; spread on thumb, index, middle	Tendon pulls, spring returns; built on InMoov	Designed specifically to reproduce sign-language hand poses. Best reference for which joints to add.
Robot Nano Hand	11 DOF, side-to-side knuckle motion, wrist pitch only	10 × SCS0009 + 1 × SCS15 serial-bus servos	Most dexterous, but budget is about €400–600 and it uses serial-bus servos instead of a PCA9685.

Electronics (recommended starting point)

Part	Qty	Why
ESP32 dev board	1	Receives pose commands over USB serial

Part	Qty	Why
PCA9685 16-channel PWM board	1	Drives up to 16 servos over I2C with 2 pins
Finger servos	5–6	Size set by the hand design. MG90S gives 1.8–2.2 kg·cm, which may be too weak for tendon pulls; InMoov uses larger servos
Thumb rotation servo	1	Needed for A, S, T, E, O and number handshapes
Wrist rotation servo	0–1	Unlocks orientation letters (G, H, K, P) and J/Z motion
5–6 V bench or brick supply	1	Sized after measuring real stall current; never power servos from the ESP32
Braided fishing line, springs, M3 screws, capacitor across servo rail	—	Tendons, returns, and brown-out protection

Open hardware questions: final servo model and torque, total stall current, whether finger spread is added for U/V, and printer time per part.

Software and AI pipeline

The software is one Python app with four stages and a browser dashboard. It runs fully in simulation mode, so the software team can build and test before the hand exists.

Stage	Recommended tool	Alternatives to research	Output
Speech-to-text	faster-whisper , <code>small</code> or <code>base</code> model, int8 on CPU (MIT licence, up to 4× faster than openai/whisper)	Browser Web Speech API, Vosk (offline), cloud Whisper API	<code>"hello my name is sam"</code>

Stage	Recommended tool	Alternatives to research	Output
AI gloss step	LLM (Claude or GPT) with a strict system prompt and JSON output	Rule-based keyword matcher as fallback and baseline	<pre>{"gloss": ["HELLO", "NAME", "FS: SAM"]}</pre>
Sequencer	Python: look up keyframes, add 250–400 ms transitions, hold each sign	—	Timed list of poses
Transport	pyserial over USB	WebSocket / MQTT over Wi-Fi (stretch goal)	One JSON line per pose
Dashboard	FastAPI + WebSocket + a web page with a 2D/3D hand	Streamlit	Live transcript, gloss, current sign, virtual hand

How the AI step works

1. The system prompt lists every supported sign ID and every letter the hand can fingerspell.
2. The model drops words ASL does not sign (articles, "is", "are"), maps meaning to supported signs, and marks unsupported words as **FS:** (fingerspell) only if every letter is supported.
3. The output is validated in code: any ID not in the library is rejected, so the model cannot invent signs.
4. A rule-based matcher runs the same job without an LLM. It is the offline fallback and the baseline you compare the LLM against in the report.

Why this split matters for marks: the LLM is the Generative AI piece, the validation layer is the responsible-AI piece, and the baseline comparison is the evaluation piece.

Interface contract

Both teams agree on two things in week 1: the sign library format and the serial message. After that, hardware and software can change independently.

Sign library (`signs.json`) — every joint is a normalised value from 0.0 (open / neutral) to 1.0 (fully closed / fully rotated). No servo angles appear here.

```
{
  "A": {
    "type": "static",
    "validated_by_signer": false,
    "frames": [
      { "thumb": 0.2, "thumb_rot": 0.0, "index": 1.0, "middle": 1.0,
        "ring": 1.0, "pinky": 1.0, "wrist": 0.5, "hold_ms": 800 }
    ]
  },
  "J": {
    "type": "motion",
    "requires": ["wrist"],
    "frames": [
      { "thumb": 0.6, "thumb_rot": 0.4, "index": 1.0, "middle": 1.0, "ring":
1.0, "pinky": 0.0, "wrist": 0.5, "hold_ms": 150 },
      { "thumb": 0.6, "thumb_rot": 0.4, "index": 1.0, "middle": 1.0, "ring":
1.0, "pinky": 0.0, "wrist": 1.0, "hold_ms": 400 }
    ]
  }
}
```

Serial message (laptop → ESP32) — one JSON line per pose, newline-terminated:

```
{"t":"pose","id":"A","j":[0.2,0.0,1.0,1.0,1.0,1.0,0.5],"ms":300}
```

`j` is the joint array in fixed order (thumb, thumb_rot, index, middle, ring, pinky, wrist); `ms` is the transition time. The ESP32 replies `{"ok":"A"}` when the move finishes, which the dashboard uses for timing.

Calibration (`calibration.json` , owned by hardware) — per joint: PCA9685 channel, angle at 0.0, angle at 1.0, and an invert flag. Re-calibrating never touches the sign library.

Target sign list

Tier 1 is the promise; Tiers 2 and 3 depend on which joints the hand gets. Every sign stays "draft" until a fluent signer confirms it.

Tier	Signs	Joints needed	Notes
1 — must have	Numbers 1, 2, 3, 4, 5	5 finger bends + thumb bend	ASL 3 uses thumb, index, middle
1 — must have	I LOVE YOU	Same	Thumb, index, pinky extended
1 — must have	Letters A, B, C, D, E, F, I, L, O, S, W, Y	5 bends + thumb rotation	Thumb rotation separates A / S / E and closes O, D, F
2 — should have	NO (index + middle close onto thumb)	Same, animated	A finger-only motion sign; good showcase of keyframes
2 — should have	Numbers 0, 6, 7, 8, 9	Thumb opposition to each finger	Depends on how far the thumb can cross the palm
2 — should have	Fingerspelled words using Tier 1 letters (e.g. names)	—	The AI only fingerspells words whose letters are all supported
3 — stretch	U vs V, K, R	Finger spread / crossing	Needs an abduction joint (see PARLOMA design)
3 — stretch	G, H, P, Q	Wrist rotation	Orientation-dependent letters
3 — stretch	J, Z, YES	Wrist rotation / flexion + motion	J and Z are traced; YES is an S-hand nod

That gives about 18 Tier 1 signs, which is enough to spell short words like "BOLD", "CODE" or "SAIL" and to answer with numbers.

Research tasks (this phase)

The research phase should end with three locked decisions: the hand design and joint count, the Tier 1 sign list, and the speech/AI stack. Each task below feeds one of them.

Hardware (arm lead)

- ☐ Compare InMoov, PARLOMA and Robot Nano Hand; pick one and list its joints
- ☐ Decide on a second thumb servo, wrist rotation and finger spread

- ☐ Pick servos from the design's torque needs; find real stall current and size the supply
- ☐ Estimate print time and filament per part; confirm printer access at SAIT
- ☐ Price the full bill of materials

Sign language (whole team)

- ☐ Learn the ASL manual alphabet and numbers 0–10 from a Deaf-led source
- ☐ Photograph each Tier 1 handshape from the front and side as a reference
- ☐ Read two or three sources on why Deaf communities have criticised sign-language gloves and robots
- ☐ Contact SAIT Accessibility Services and a Calgary Deaf organisation to find a fluent signer for review

Software and AI (your side)

- ☐ Test faster-whisper `base` and `small` on a laptop CPU: word accuracy and seconds per sentence
- ☐ Draft the gloss prompt; test it on 30 sample sentences
- ☐ Build the rule-based baseline matcher
- ☐ Choose the dashboard stack and 3D hand approach
- ☐ Confirm the ESP32 + PCA9685 Arduino library and a serial test sketch

Course and ethics

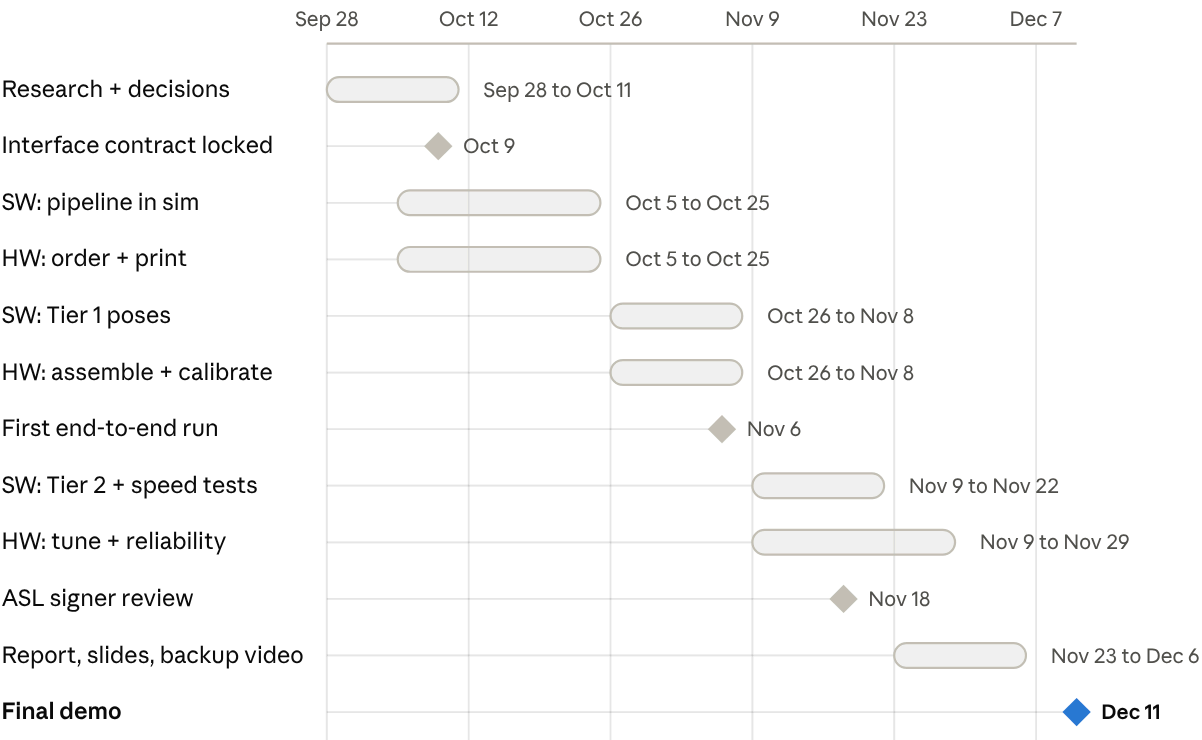
- ☐ Confirm deliverables and marking rubric with the instructor
- ☐ Check whether testing with a volunteer signer needs any SAIT approval
- ☐ Collect sources for the report (see References)

Timeline and roles

Hardware and software run in parallel for five weeks and meet at the first end-to-end run on Nov 6. The Dec 11 demo date is a placeholder until the real deadline is confirmed.

Hand and software meet on Nov 6; final demo by Dec 11

Two-week ticks from Sep 28, 2026. Demo date assumed until the instructor confirms it.



SIGNOVA timeline · fall 2026

Role	Owns	Key hand-off
Project lead / AI + software (you)	Speech-to-text, AI gloss, sequencer, dashboard, ESP32 firmware skeleton	Sign library + serial protocol by Oct 9
Hardware lead (friend)	Hand design, printing, servos, wiring, power, calibration file	Calibrated hand by Nov 6
Sign-language + ethics owner	Reference photos, signer outreach, validation sessions, ethics section	Signer review by Nov 18
Report + presentation owner	Report, slides, demo script, backup video	Draft by Nov 30

Responsible AI and ethics

The main ethical risk is claiming more than the prototype does. Hearing-led sign-language gadgets have been widely criticised by Deaf people; SIGNOVA should learn from that and say so in the report.

What the literature says ([Machine translation of sign languages, Wikipedia](#))

- Sign languages use "the hands, arms, head, shoulders, torso, and parts of the face", so hand shapes alone are not ASL.
- The SignAloud gloves were criticised because they did "not incorporate facial expressions and other nonmanual markers" and gave only one-way communication.
- Their hearing creators later said they were working more closely with ASL users "rather than the assumed needs".
- Still to read for the report (could not be opened here): [Do deaf communities actually want sign language gloves?](#) and [ASHA Leader on sign-language gloves](#).

What SIGNOVA commits to

1. **Honest framing:** "a research prototype exploring physical AI", never "a translator" or a replacement for interpreters.
2. **Deaf input:** at least one fluent signer reviews every sign; their feedback, including negative feedback, goes in the report. Offer thanks or compensation for their time.
3. **Constrained AI:** the model can only output signs in the validated library; unknown words are fingerspelled or skipped, never guessed.
4. **Privacy:** audio is processed in memory and not stored; a notice on the dashboard says so. If a cloud API is used, say which one.
5. **Transparency:** the dashboard shows the gloss and marks unvalidated signs, so viewers see what the AI decided.
6. **Future work framing:** list what Deaf users said they would actually want, not what the team assumed.

Evaluation plan

SIGNOVA is judged on four numbers, each measured the same way before and after tuning.

Metric	How it's measured	Target
Sign recognition	Fluent signer watches each Tier 1 sign in random order and names it, without the dashboard	$\geq 80\%$ correct
End-to-end latency	Time from speaker stopping to first finger movement, 20 sentences, logged by the app	≤ 3 s median
Gloss accuracy	30 test sentences; LLM output vs. a hand-written reference gloss; compared with the rule-based baseline	LLM beats baseline; 0 invented signs
Repeatability	Same 10-sign sequence run 20 times; count missed or wrong poses and hardware faults	0 faults, ≤ 1 wrong pose
Speech accuracy	Word error rate on the 30 sentences, quiet room	Reported, no target

Record every trial in a CSV from the start; the report's results section is then just charts over that file.

Risks and mitigations

The biggest risks are mechanical, so the plan keeps the software demo-able even if the hand fails.

Risk	Mitigation	Owner
Hand can't form signs clearly (too few joints, weak servos)	Lock joint count in research; pick Tier 1 signs the hand can make; test one finger before printing all five	Hardware
Servos brown out or ESP32 resets on demo day	Separate supply sized to stall current, capacitor on servo rail, never power servos from USB	Hardware
Tendons stretch or slip	Braided line, not nylon; tension screws; re-calibrate before demo	Hardware

Risk	Mitigation	Owner
Speech recognition fails in a noisy room	Push-to-talk, close mic, typed-input fallback on the dashboard	Software
LLM invents signs or is slow / offline	Output validated against the library; rule-based fallback; local option	Software
No ASL signer available in time	Start outreach in week 1; ask SAIT Accessibility Services and Calgary Deaf organisations; report honestly if review is limited	Sign-language owner
Integration slips past Nov 6	Simulation mode means software is finished independently; backup video recorded early	Whole team

Demo and presentation plan

The live demo shows one sentence travelling through every stage, with the dashboard on a projector and the hand beside it.

1. **Hook (30 s):** someone says "Hello, I love you" → hand signs I LOVE YOU. Dashboard shows transcript, gloss, and which words were dropped.
2. **Fingerspelling (30 s):** say a short name built from Tier 1 letters; hand spells it letter by letter.
3. **Under the hood (60 s):** show the gloss JSON, the validation step rejecting an invented sign, and the rule-based baseline.
4. **Honesty slide (30 s):** what the hand can't do (facial grammar, two hands, movement), and what the ASL reviewer said.
5. **Future work (30 s):** more joints, two hands, other sign languages, Deaf-led design.

Backups: typed input if the mic fails; simulation-only mode if the hand fails; a recorded video of the full run.

Browser prototype: an interactive mock-up of the dashboard and virtual hand is at [SIGNOVA Lab](#), for early testing of the gloss idea and for the pitch.

References

- [InMoov — Hand and Forearm](#)
- [Bulgarelli et al. \(2016\), A Low-Cost Open Source 3D-Printable Dexterous Anthropomorphic Robotic Hand with a Parallel Spherical Joint Wrist for Sign Languages Reproduction](#)
- [Robot Nano Hand](#)
- [MG90S servo specifications \(Components101\)](#)
- [faster-whisper \(SYSTRAN, GitHub\)](#)
- [Machine translation of sign languages \(Wikipedia\)](#)
- To read: [Do deaf communities actually want sign language gloves?](#)
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